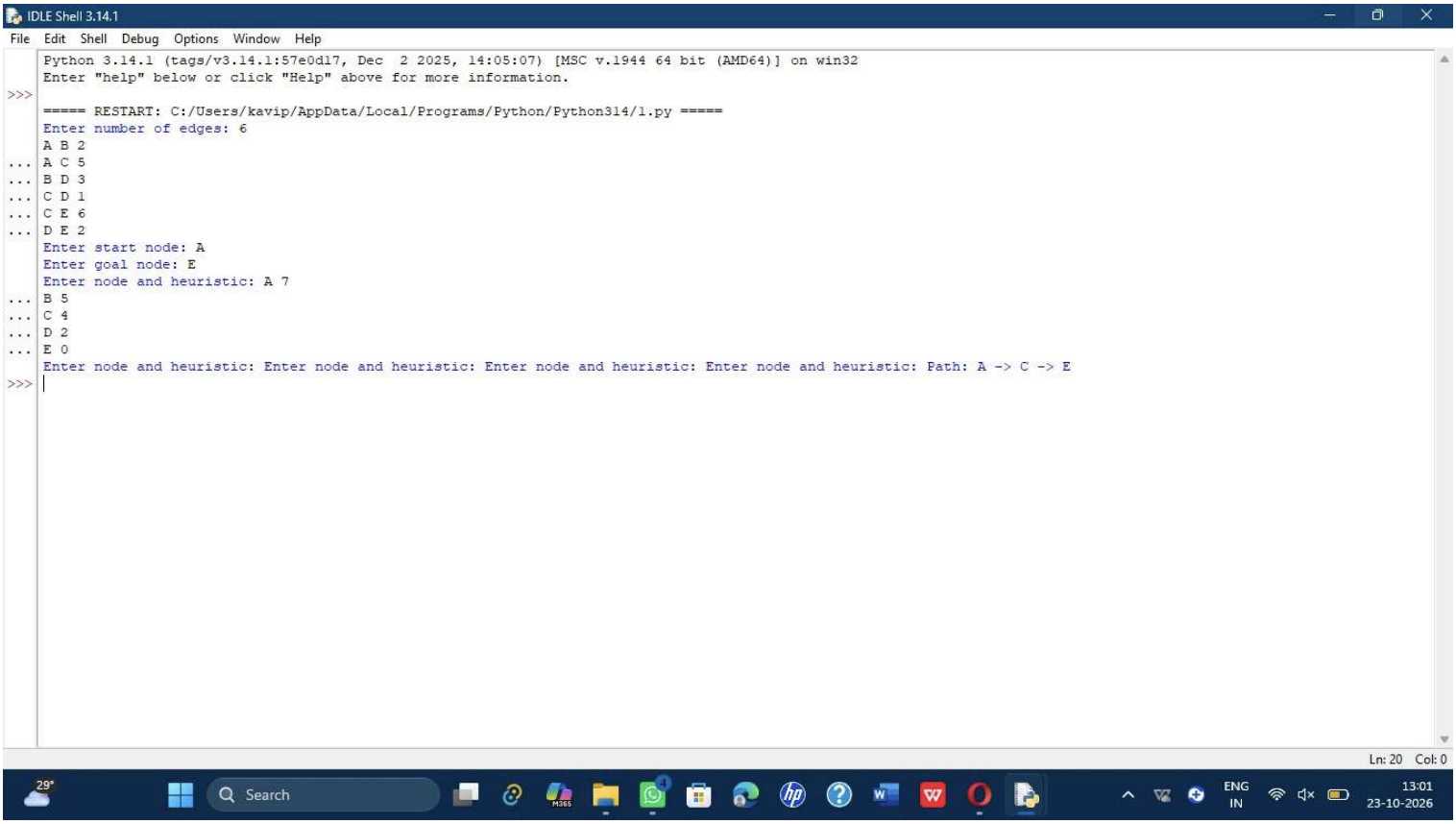

IDLE Shell 3.14.1
File Edit Shell Debug Options Window Help
Python 3.14.1 (tags/v3.14.1:57e0d17, Dec 2 2025, 14:05:07) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/kavip/AppData/Local/Programs/Python/Python314/1.py =====
Enter number of edges: 6
A B 2
... A C 5
... B D 3
... C D 1
... C E 6
... D E 2
Enter start node: A
Enter goal node: E
Enter node and heuristic: A 7
... B 5
... C 4
... D 2
... E 0
Enter node and heuristic: Enter node and heuristic: Enter node and heuristic: Enter node and heuristic: Path: A -> C -> E
>>>
Ln: 20 Col: 0
29°
Search
ENG
IN
13:01
23-10-2026

1.py - C:/Users/kavip/AppData/Local/Programs/Python/Python314/1.py (3.14.1)

File Edit Format Run Options Window Help

```
    cost = int(cost)

    if u not in graph:
        graph[u] = []
    if v not in graph:
        graph[v] = []

    graph[u].append((v, cost))

start = input("Enter start node: ")
goal = input("Enter goal node: ")

heuristic = {}

for node in graph:
    nl, h = input("Enter node and heuristic: ").split()
    heuristic[nl] = int(h)

pq = [(heuristic[start], start, [start])]
visited = set()

while pq:
    h, node, path = heapq.heappop(pq)

    if node == goal:
        print("Path:", " -> ".join(path))
        break

    if node in visited:
        continue

    visited.add(node)

    for neighbour, cost in graph[node]:
        if neighbour not in visited:
            heapq.heappush(
                pq,
                (heuristic[neighbour], neighbour, path + [neighbour])
            )
else:
    print("No path found")
```

Ln: 50 Col: 0

